

Lecture 7 Summary

This lecture covers what clustering is, how to measure distance between data points, and how hierarchical clustering works, it also introduces Mahalanobis distance.

What Is Clustering?

Clustering finds groups of objects that are like each other inside a group and different from objects in other groups. Good clustering has small differences inside each cluster and large differences between clusters.

Every clustering task needs three things:

- The data to be clustered.
- A way to measure similarity or distance.
- A clustering algorithm, which is the procedure that does the grouping.

Distance Measures

Minkowski distance is a general form of Euclidean distance. A parameter r changes its type:

- $r = 1$ gives Manhattan (city block) distance. It adds up the differences in each dimension. Hamming distance is an example.
- $r = 2$ gives the usual Euclidean distance.
- $r \rightarrow \infty$ gives supremum distance. It is the largest difference in any single dimension.

Other options include cosine similarity, correlation, and Mahalanobis distance. The best choice depends on the data.

Hierarchical Clustering

This method builds a tree of nested clusters. The tree is represented as a dendrogram, which maintains the order of the merges. It is useful for meaningful structures like groups of genes, phylogenetic trees, or the animal kingdom.

The steps are:

1. compute the distance (or similarity) matrix
2. Each data point is considered a cluster.
3. Merge the two closest clusters.
4. Update the distance matrix.
5. Repeat steps 3 and 4 until only one cluster is left.

The update step needs a rule for the distance between clusters. The options are:

- Min: the closest pair of points.
- Max: the farthest pair of points.
- Group average.
- Distance between centroids.

Gene Expression Example

The lecture clusters five genes measured in a wild type and three mutants. It uses correlation as the similarity measure, because the shape of the expression pattern matters more than the raw size of the values. The merge order was:

1. At2g34930 and At1g30720 merge first (correlation 0.9733 after updating).
2. At4g35770 joins them.
3. At2g05540 joins next.
4. At4g27450 joins last. It is negatively correlated with the others, so it is the most different gene.

The lecture links to scikit-learn for hierarchical clustering in Python and recommends a gene expression clustering job for Project 2. And the task covers data cleaning, building a distance matrix, and running the clustering.

Mahalanobis Distance

Mahalanobis distance measures distance while taking the data distribution into account. It uses the covariance matrix.

The lecture gives a quiz example. A 1-point difference on a quiz with standard deviation 10 is small. The same 1-point difference in a quiz with standard deviation 1 is large. A fair comparison counts how many standard deviations apart from two points are.

The lecture also shows min-max normalization. It rescales each variable, such as height and weight, to a range from 0 to 1.